INCH-POUND
MIL-PRF-32645
28 January 2020

PERFORMANCE SPECIFICATION

WATER CLOSET, VITREOUS CHINA, GRAVITY-FED, FOR SURFACE SHIPS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers U.S. Navy shipboard vitreous china, gravity-fed water closets (WC) on surface ships where seawater is used for flushing.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

OPNAV INSTRUCTIONS

OPNAVINST 4700.7 - Maintenance Policy for Navy Ships

(Copies of this document are available online at https://www.secnav.navy.mil/doni/default.aspx.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4510

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:18Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.18.2/CSA B125.2	-	Plumbing Waste Fittings
ASME A112.19.2/CSA B45.1	-	Ceramic Plumbing Fixtures (with 10/18 Errata)
ASME A112.19.5/CSA B45.15	-	Flush Valves and Spuds for Water Closets, Urinals, and Tanks
ASSE 1002/ASME A112.1002/CSA B125.12	-	Anti-Siphon Fill Valves for Water Closet Tanks

(Copies of these documents are available online at www.asme.org.)

CSA GROUP STANDARDS

CSA B125.3 - Plumbing Fittings

(Copies of this document are available online at www.csagroup.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Design. Unless otherwise specified (see 6.2), fixtures shall be constructed in accordance with the gravity flush design and material details of ASME A112.19.2/CSA B45.1. Fixtures shall be designed with minimal construction, installation, and maintenance requirements. Fixtures shall be designed for floor-mounted and rear-outlet use as specified (see 6.2).

3.2.1 Access. Standard fixtures shall pass through a 26-inch (660-millimeter) wide by 66-inch (1676-millimeter) height shipboard hatch with 6.5-inch (165-millimeter) radius corners without major disassembly. When required (see 6.2), fixtures may be of the modular type, capable of being disassembled and reassembled at the location of installation.

3.2.2 Weight. Maximum weight shall not exceed 75 pounds.

3.2.3 Glazing. Glaze shall be thoroughly fused to the body of the plumbing fixture. Fixtures shall have all exposed surfaces glazed in accordance with ASME A112.19.2/CSA B45.1.

3.2.4 Waste fitting openings and drainage. Unless otherwise specified (see 6.2), fixtures shall have waste outlets at the lowest point of the fixture and drain to the waste outlet in accordance with ASME A112.19.2/CSA B45.1. Waste fittings shall be compatible with shipboard plumbing fixtures. All factory-supplied plumbing waste fittings shall be in accordance with ASME A112.18.2/CSA B125.2.

3.2.5 Gravity flush tanks. Gravity flush tanks shall include an anti-siphon fill valve in accordance with ASSE 1002/ASME A112.1002/CSA B125.12 or CSA B125.3 and a flush valve in accordance with ASME A112.19.5/CSA B45.15 or CSA B125.3. Gravity flush tanks shall have provisions for overflow.

3.2.6 Water inlet openings. The water inlet openings shall be arranged to rinse all exposed internal surfaces.

3.2.7 Color. Fixtures shall be white in color.

3.2.8 Water consumption. Water consumption shall not exceed 1.6 gallons per flush.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:18Z
Check the source to verify that this is the current version before use.

3.2.9 Rim profiles. Unless otherwise specified (see 6.2), water closet rim profiles shall be in accordance with ASME A112.19.2/CSA B45.1.

3.3 Additional requirements.

3.3.1 Integral trap diameter. Fixtures shall have integral trap diameters sized in accordance with ASME A112.19.2/CSA B45.1. Bowl traps shall have a diameter that allows a solid ball with a minimum diameter of 1.5 inches (38 millimeters) to pass.

3.3.2 Interior dimensions. Bowls shall have a minimum water surface dimension of 5 × 4 inches (125 × 102 millimeters) when measured on a flat and level surface in accordance with ASME A112.19.2/CSA B45.1 (see 4.4).

3.3.3 Exterior dimensions. Fixtures shall have a maximum rim height of 13.5 inches (343 millimeters) in accordance with ASME A112.19.2/CSA B45.1. Fixtures shall have a maximum exterior length of 25 inches (635 millimeters) and a maximum exterior width of 15 inches (381 millimeters).

3.3.4 Fill valve opening diameter and location. The fill valve shall be the pilot valve type only. The fill valve opening shall have a diameter of 1.1 inches (28 millimeters) and shall be in accordance with ASME A112.19.2/CSA B45.1.

3.3.5 Critical level. The critical level mark on the fill valve shall be at least 1 inch (25 millimeters) above the flush tank overflow.

3.3.6 Spuds. Fixtures shall include appropriately sized spuds that are in accordance with ASME A112.19.2/CSA B45.1. Spud fittings and gaskets shall be furnished with water closets.

3.3.7 Materials and construction. Fixtures shall be constructed with materials and through processes in accordance with ASME A112.19.2/CSA B45.1.

3.3.8 Non-magnetic materials. Fixtures shall be furnished with non-magnetic parts as specified (see 6.2).

3.3.9 Condensation. Fixtures shall incorporate a condensation-free (insulated) tank in accordance with ASME A112.19.2/CSA B45.1.

3.4 Performance characteristics.

3.4.1 Flushing devices used with fixtures. Flushing devices shall deliver water at a sufficient rate and quantity in accordance with ASME A112.19.2/CSA B45.1.

3.4.2 Shipboard environment. Fixtures shall be able to withstand type I vibration requirements in accordance with MIL-STD-167-1 without mechanical damage, loss of functionality, cracking, or leakage.

3.5 Installation characteristics.

3.5.1 Installation instructions. Installation instructions shall be as specified (see 6.2).

3.5.2 Mounting. Fixtures shall be configured as the floor-mounted type. Deck pad and mounting hardware shall be provided by the installing activity.

3.5.3 Roughing-in details. Fixture outlets shall rough-in at 10, 12, or 14 inches (254, 305, or 356 millimeters) in accordance with ASME A112.19.2/CSA B45.1 or as specified (see 6.2).

3.6 Clearances. Fixtures shall be able to operate with minimal clearances required between the unit and adjacent equipment or structures. Allowable range of clearance shall be 0 to 3 inches (0 to 77 millimeters).

3.7 Maintenance. Fixtures shall be maintainable, repairable, and upgradeable by shipboard personnel, given installation requirements (see 3.5). Individual components shall be replaceable at shipboard operational-level as defined in OPNAVINST 4700.7. Maintenance and repair access shall be from the front of the unit. Components shall be able to be quickly disconnected for ease of repair or replacement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:18Z
Check the source to verify that this is the current version before use.

3.8 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete assembly when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.3.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 (see table I).

4.3.1 Lot. A lot shall consist of all units of the same type that are produced under similar conditions and are ready for inspection or shipment at one time. Unless otherwise specified (see 6.2), the lot size shall be expressed in terms of the number of finished fixtures.

4.3.2 Sampling for conformance inspection. Samples shall be randomly selected from each lot. At a minimum, the sampling rate shall be 10 percent on lots of 50 or less, and 5 percent on lots greater than 50.

TABLE I. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Design	X	X	3.2	4.4
Trap diameter	X	X	3.3.1	4.4
Interior dimensions	X	X	3.3.2	4.4
Exterior dimensions	X	X	3.3.3	4.4
Fill valve diameter and location	X	X	3.3.4	4.4
Critical level	X	X	3.3.5	4.4
Spuds	X		3.3.6	4.5.2
Materials and construction	X		3.3.7	4.5.2
Non-magnetic materials	X		3.3.8	4.5.2
Condensation	X		3.3.9	4.5.2
Flushing devices used with fixtures	X		3.4.1	4.5.3
Environmental conditions	X		3.4.2	4.5.1
Installation	X	X	3.5	4.4
Clearances	X		3.6	4.5.3
Maintenance	X	X	3.7	4.4
Recycled, recovered, environmentally preferable, or biobased materials	X	X	3.8	4.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:18Z
Check the source to verify that this is the current version before use.

4.4 Examination. Fixtures shall be subjected to a thorough inspection to confirm adherence to design characteristics as identified in 3.2, construction and installation requirements as identified in 3.2 and 3.5, dimensional requirements as identified in 3.3.1, 3.3.2, 3.3.3, 3.3.4, and 3.3.5, maintenance requirements as identified in 3.7, and material requirements established in 3.8. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of verification.

4.5.1 Shipboard environment. Fixtures shall be tested under normal operating conditions in accordance with MIL-STD-167-1, type I equipment. Frequency testing shall be in the range of 4 to 25 Hz. Fixtures shall be secured to the test machine in the same manner that they will be secured on shipboard. Fixtures may be tested and qualified with or without isolation mounts. Failure of the unit to perform its function during or after testing shall constitute failure of this test. Testing shall be performed to verify adherence to shipboard environment requirements established in 3.4.2.

4.5.2 Materials, finishes, structural integrity, and seals. Adherence to the design requirements established in 3.3.6, 3.3.7, 3.3.8, and 3.3.9 shall be verified using test methods in accordance with ASME A112.19.2/CSA B45.1.

4.5.3 Operational compliance. Fixtures shall conform to test methods in accordance with ASME A112.19.2/CSA B45.1. Operational testing shall be performed to verify water consumption as identified in 3.2.8, flushing devices as identified in 3.4.1, and operational clearances as identified in 3.6.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This product is intended for use on U.S. Navy surface ships to provide a reliable gravity-fed water closet for use in sanitary spaces. A gravity-fed water closet is defined as a vessel that stores a predetermined quantity of water and includes a flushing device to discharge water (plus some through-flow from the water supply line) into a water closet bowl. This product is not intended for use onboard submarines or for new construction surface combatants and aircraft carriers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is required (see 3.1).
c. Alternative design requirements (see 3.2).
d. Mounting requirements (see 3.2 and 3.5.2).
e. Modular construction requirements if needed (see 3.2.1).
f. Waste fitting openings, and drainage requirements if other than specified (see 3.2.4).
g. Non-standard rim profiles (see 3.2.9).
h. Non-magnetic material requirements (see 3.3.8).
i. Installation instruction requirements (see 3.5.1).
j. Roughing-in details requirements (see 3.5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:18Z
Check the source to verify that this is the current version before use.

k. Lot size (see 4.3.1).

l. Packaging requirements (see 5.1).

m. Data requirements (see 6.3).

6.3 Data requirements. When specified (see 6.2), illustrations should be provided detailing the installation methods of water closets. Illustrations should be provided showing the disassembly and reassembly of water closets if fixtures are of the modular type.

6.4 Subject term (key word) listing.

Device, flushing
Fixture
Glaze
Integral
Plumbing
Spud
Toilet
Trap
Trunk, gravity, flush

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 03

Review activities:
Army – MI
Navy – AS, YD
DLA – CQ
GSA – FAS

Preparing activity:
Navy – SH
(Project 4510-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:18Z
Check the source to verify that this is the current version before use.